Batteries of the Future: What will make your gadgets run like Duracell bunnies?

You're on a really important phone call or writing a really important email and your battery dies. Don't you feel like your world has come to a standstill and that your life is over?

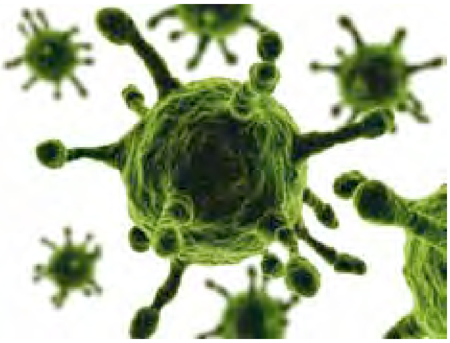
Raj Saxena
readersletters@thinkdigit.com

Batteries dictate the world of technology. All the gadgets in the world are pieces of junk without power. Hardware such as smartphones, laptops and tablets are getting faster and more capable while also shrinking in size. But the extra room gained inside devices is often used for more components and features, rather than bigger batteries which just make the devices heavier – an undesirable aspect. With our growing addiction to mobility, it's assuring to know about on-going research in battery technology. Research that will enable products to run longer.

Nature builds batteries

"My dream: to be able to drive a virus-powered car," says Angela Belcher who holds a PhD in inorganic chemistry from University of California. She looked towards nature for inspiration to solve our current energy problems. Currently the head of the Biomolecular group at MIT, her research brings together the three fields of Molecular Biology, Electrical Engineering and Material Chemistry.

Such a battery would be "alive" as viruses would be doing the heavy lifting of taking the electrons through the electrolyte. These viruses were successfully developed by the MIT team around three years ago. This is the first time that viruses have been genetically engineered to build both, the positively and negatively charged ends of a battery. The new



Reasearch is on to use viruses as agents which transfer electrons in a battery

technology produces the same amount of power as the state-of-the-art batteries used in hybrid cars today. These batteries can be manufactured by a cheap, environmentally-safe process. The process of development takes place at room temperature and needs no environmentally unfriendly solvents – something that hasn't been achieved by the other battery-tech gurus. Experts say that the anode is simpler to develop in a virus battery as compared to the cathode, the reason being that it needs to be highly conducting to be a fast electrode. Therefore, the current research is focussed on building a highly powerful cathode to pair up with the anode.

Graphene Power

Next, we come to Graphene, a material that has been under intensive research in recent times due to its versatility. In fact, IBM's breathing batteries face stiff competition from the graphene batteries. Graphene is a material that consists of a layer of carbon that is only one atom thick.

IBM researchers believe that graphene isn't stable enough for long-term usage but researchers led by Zihan Xu of the Department of Applied Physics and Materials Research Centre at Hong Kong Polytechnic University disagree. The team has developed graphene batteries by attaching gold and silver electrodes to a graphene sheet. This assembly was immersed in copper chloride solution and an output of 0.35V was measured. An assembly of six such cells in a series was connected to a Light Emitting Diode (LED). The device generated the same amount of power for about 25 days, but after a month it dropped to 40 mV.

In a research paper, Professor Xu mentions that the output of the battery increases when the electrolyte (copper chloride solution) is heated as well as when the assembly is exposed to ultrasound energy. This is because the kinetic energy of the molecules increases when

A graphene battery involves attaching Gold and silver electrodes to a graphene sheet